

Lu Zhang has written a series of papers explaining the value premium in terms of how value firms are risky as a result of their underlying production technologies. An important paper is Zhang (2005), which builds on the *production-based asset pricing* framework introduced by Cochrane (1991, 1996). Cochrane teaches us to look at firm investment to study firm returns. The gist of the Cochrane–Zhang story is as follows. Value firms and growth firms differ in how flexible they are and how quickly they can respond to shocks. During bad times, value firms are risky because they are burdened with more unproductive capital. Think of value firms as making stodgy widgets, and when a bad time comes, they cannot shift their firm activities to more profitable activities—they are stuck making widgets. They wish to cut back on capital, but they cannot sell their specialized widget-manufacturing equipment. In economists’ jargon, they have *high and asymmetric adjustment costs*. Growth firms, however, can easily divest because they employ hotshot young employees and the great bulk of their capital is human capital, not stodgy widget-making factories. Thus, value firms are fundamentally riskier than growth firms and command a long-run premium.

Rational Implications for Asset Owners

The literature is still debating whether the bad times defined by these theories are truly bad times. But this academic debate is not that relevant to an asset owner; bickering over a Cochrane–Zhang story versus a story about another risk factor is not what the asset owner contemplating a value tilt should be doing.

Remember that the average investor holds the market portfolio. The asset owner should take these rational theories and ask, given that each factor defines a different set of bad times, are these actually bad times for me? If I do not need to eat less during periods when investment growth is low, then this is not as bad a time for me as it is for the average investor. Thus, I have a comparative advantage in holding value stocks and can harvest the value premium. Other investors are not comfortable holding value stocks (and should hold growth stocks instead) because they cannot afford to shoulder the losses generated by value stocks during bad times. Overall, the average investor holds the market even though some investors prefer value stocks and some investors prefer growth stocks. Which type you are—value or growth—depends on your own behavior during each of these bad times.

3.5. BEHAVIORAL THEORIES OF THE VALUE PREMIUM

Most behavioral theories of the value premium center around investor *overreaction* or *overextrapolation* of recent news. The standard story was first developed by Lakonishok, Shleifer, and Vishny (1994). Investors tend to overextrapolate past growth rates into the future. The posterchild growth stock example at the time of writing is Apple Inc. (AAPL), which has achieved tremendous growth over the last few years by introducing a series of must-have products. Investors mistake Apple’s past high growth for future high growth. Growth firms, in general, have